

Data Guide for Coppola, Maggiori, Neiman, and Schreger: “Redrawing the Map of Global Capital Flows”

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The following is a list of the raw input files required to run the replication code for Coppola et al. (2021), together with short descriptions of the data. All paths are relative to the root data folder referred to as `DATA_PATH` in the code. Please see the accompanying README file for more details on how these data are deployed. Files marked with [*] are publicly available and are included in the accompanying replication packet. Files not marked with [*] must be purchased from the respective data providers; in these cases the list below provides information as to how the data can be sourced. For some of these files, in the cases in which the structure of the data might be ambiguous, the replication packet includes sample version of the files that only include sample rows — these files are included in order to demonstrate the structure of the data. We indicate below the cases in which we include such mock files. The path structure outlined below is the one that is referenced in the code.

1. CUSIP Global Services (CGS) Data

- *Paths:*

1. `raw/cgs_master/ALLMASTER_ISSUER.PIP.zip`
2. `raw/cgs_master/INCMSTR.PIP.zip`
3. `raw/cgs_master/ALLMASTER_ISIN.PIP.gz`
4. `raw/cgs_master/AIMASTER.PIP.zip`
5. `raw/cgs_master/CPMASTER_ATTRIBUTE.PIP.zip`
6. `raw/cgs_master/CPMASTER_ISSUE.PIP.zip`
7. `raw/cgs_master/CPMASTER_ISSUER.PIP.zip`
8. `raw/cgs_master/FFAPlusMASTER.PIP.zip`
9. `raw/cgs_master/TBA Master File [Vintage].zip`
10. `raw/cgs_lei_plus/CBRLEIMSTR.PIP`

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11. raw/cgs_master/master_[Vintage].GM
12. raw/cgs_master/master_[Vintage].SB
13. raw/cgs_master/master_[Vintage].FM
14. raw/cgs_master/master_[Vintage].FD
15. raw/cgs_master/master_[Vintage].IB
16. raw/cgs_master/previous_versions/*

- *Description:* The files listed here are the security- and issuer-level master files for global CUSIP-bearing securities that are provided by CUSIP Global Services (CGS). We have included in the replication package mock versions of these files with sample rows only, to illustrate their structure. These files can be obtained commercially from CGS (cusip.com).
 - File (1) is the main issuer-level master file from CGS and provides data from the CUSIP_db and CINS_db issuer master file products.
 - Files (2) and (3) are the main security-level master files from CGS and provide data from the CUSIP_db, ISIN_db, and CINS_db issue master file products.
 - File (4) is the CGS Associated Issuers master file, which provides information about relationships among various CUSIP6 issuer numbers – this is useful in establishing which CUSIP6 numbers belong to the same issuing entities.
 - Files (5) through (7) contain data for commercial paper issuers and issues.
 - File (8) contains security-level information on 144a issues.
 - File (9) contains security-level information on TBA issues.
 - File (10) constitutes the CGS LEI Plus mapping data product by CGS, which provides information mapping CUSIPs to Legal Entity Identifiers (LEIs).
 - Files (11) through (15) contain security-level information on agency and sovranational issues: file (11) contains GNMA issues; file (12) contains SBA issues; file (13) contains FNMA issues; file (14) contains FHLMC issues; and file (15) contains World Bank issues.
 - The files under path (16) are previous vintages of the same CGS master files that we received during the course of the project. The build code consolidates these old versions and the new versions in order to catch any potential data that may have been excluded from more recent issues. However, this is not essential and the code can be easily adapted to run with only the latest version of the CGS data.

2. Morningstar Holdings Data

- *Paths:*
 1. raw/morningstar_holdings/HD_[YYYY]_y.dta
 2. raw/morningstar_holdings/[US/NonUS]_[YYYY]_m_step3.dta

- This code makes use of the the Morningstar holdings data for mutual funds and ETFs described in Maggiori, Neiman and Schreger (2020). Please see that paper for details on construction of the Morningstar holdings sample. The main files that are used in the code are referred to as “*HD*” (Holding Detail) files in the build code of MNS. In order to build a country assignment for each issuer from the Morningstar fund reports, we also make use of the files referred to as “*step3*” files in the build code of MNS. The raw holdings data can be purchased from Morningstar (morningstar.com). A copy of the MNS build code can be found at journals.uchicago.edu/doi/suppl/10.1086/705688.

3. Dealogic Data

- *Paths:*
 1. raw/dealogic/stata/Company.dta
 2. raw/dealogic/stata/CompanyListings.dta
 3. raw/dealogic/stata/CompanyNAICSCodes.dta
 4. raw/dealogic/stata/CompanySICCodes.dta
 5. raw/dealogic/stata/DCMDeal.dta
 6. raw/dealogic/stata/DCMDealTranches.dta
 7. raw/dealogic/stata/DCMDealTranchesISINs.dta
 8. raw/dealogic/stata/DCMDealTranchesIssueCharacteristics.dta
 9. raw/dealogic/stata/DCMDealTranchesProceeds.dta
 10. raw/dealogic/stata/DCMDealTranchesValue.dta
 11. raw/dealogic/stata/DCMIssueCharacteristics.dta
 12. raw/dealogic/stata/DCMSecurityType.dta
- *Description:* All files correspond to tables in the Dealogic Debt Capital Markets (DCM) database, which can be purchased directly from Dealogic (dealogic.com). The Dealogic data is usually delivered in SQL or SAS format, and should be converted to Stata format using software such as StatTransfer.

4. NAIC Holdings Data, Provided by S&P Global

- *Paths:*
 1. raw/sp_insurance/life/[Firm]-[YYYYQQ].xlsm
 2. raw/sp_insurance/health/[Firm]-[YYYYQQ].xlsm
 3. raw/sp_insurance/pc/[Firm]-[YYYYQQ].xlsm
- *Description:* These files contain the quarterly security-level holdings of U.S. insurers, which are reported in regulatory reports to the National Association of Insurance Commissioners (NAIC). We use the version of the data provided by the division of S&P Global formerly known as SNL Financial. The data come in the form of *.xlsm* (Excel) files, one per insurer-quarter pair, and cover all three insurance segments in the United States: life insurers, health insurers, and property and casualty (P&C) insurers. The data can be purchased directly from S&P (spglobal.com).

5. Norwegian Sovereign Wealth Fund Holdings [*]

- *Paths:*
 1. raw/Norway_SWF/EQ_[YYYY]_Country.xlsx
 2. raw/Norway_SWF/FI_[YYYY]_Country.xlsx
 3. raw/Norway_SWF/RE_[YYYY]_Country.xlsx
- *Description:* These files contain the position-level holdings of the Norwegian sovereign wealth fund (SWF). The data is publicly available (nbim.no/en/the-fund/investments). The files starting in “EQ_” cover equity positions, while the files starting in “FI_” cover fixed income positions, and those starting in “RE_” cover real estate positions.

6. Factset Data

- *Paths:*
 1. raw/Factset/fds_stata/country_map.dta
 2. raw/Factset/fds_stata/country_map.dta
 3. raw/Factset/fds_stata/dcs_cusip_hist.dta
 4. raw/Factset/fds_stata/dcs_details.dta
 5. raw/Factset/fds_stata/ent_entity_affiliates.dta
 6. raw/Factset/fds_stata/ent_entity_coverage.dta
 7. raw/Factset/fds_stata/ent_entity_names.dta
 8. raw/Factset/fds_stata/ent_entity_structure.dta
 9. raw/Factset/fds_stata/gr_item
 10. raw/Factset/fds_stata/gr_report
 11. raw/Factset/fds_stata/sym_bbg.dta
 12. raw/Factset/fds_stata/sym_coverage.dta
 13. raw/Factset/fds_stata/sym_cusip_hist.dta
 14. raw/Factset/fds_stata/sym_cusip.dta
 15. raw/Factset/fds_stata/sym_isin_hist.dta
 16. raw/Factset/fds_stata/sym_isin.dta
 17. raw/Factset/fds_stata/sym_sec_entity.dta
 18. raw/Factset/workstation/HKG_Companies.xlsx
 19. raw/Factset/workstation/LUX_Companies.xlsx
 20. raw/Factset/workstation/IRL_Companies.xlsx
 21. raw/Factset/workstation/NLD_Companies.xlsx
 22. raw/Factset/workstation/Norway_SWF_Equity_Holdings_[YYYY]_Factset.xlsx

23. raw/Factset/workstation/vie_factset.xlsx

24. raw/Factset/workstation/china_industries_factset.xlsx

- *Description:*

- Files (1) through (17) correspond to tables in the following database products from Factset: Factset Data Management Solutions, Factset Debt Capital Structure, and Factset Geographic Revenue Exposures (GeoRev). All tables can be obtained by purchasing a subscription to these data products directly from Factset (factset.com). These tables are normally delivered via a SQL feed, and should be converted to Stata format before use.
- Files (18) through (21) contain a list of public companies and corresponding equity CUSIP codes that are confirmed to be domiciled in Hong Kong, Luxembourg, Ireland, and the Netherlands, respectively. We built these lists by using the Factset equity screening tool. Taking Hong Kong as an example, we search for companies for which both the firm’s headquarters and the firm’s ultimate parent’s headquarters are located in Hong Kong, as reported by Factset. We sort firms by market capitalization, and select the 40 largest ones that meet our criteria for inclusion in file (18). The construction of the rest of the files is analogous.
- The files under (22), one per year from 2007 to 2017, correspond to Factset’s version of the equity position-level data of the Norwegian SWF. We use these files to establish a mapping from the SWF’s equity positions to CUSIP codes. These files can be downloaded via Factset’s Excel API.¹
- File (23) contains information on a set of dynamic fields (market capitalization and equity price) and static fields (IPO date, IPO price, and shares offered at IPO) for China’s VIE companies. This data was downloaded via Factset’s Excel API. We include in the replication package an example Excel workbook that outlines the required structure and the API formulas that we used. The workbook contains three sheets, called “static”, “market_cap”, and “price”. All sheets have one row per CUSIP, and we include a single example row demonstrating the formulas. The rest of the rows should be populated with the set of equity CUSIPs in the CGS data for which the CMNS aggregation file (`cmns_aggregation.dta`, produced by the code) assigns residency in the Cayman Islands and nationality in China.
- File (24) reports the primary industry classification for the same set of CUSIP codes as in file (23). This file should have columns “cusip”, populated as above, and “industry”, populated using Factset’s Excel API. The API formula to obtain the industry classifications is `=FDS(<CUSIP_ID>,"FG_FACTSET_IND")`.

¹For example, to download the 2017 data the API formulas are the following, where “M4032427” is Factset’s identifier for the Norwegian SWF:

* *Name* column: `=FDS("M4032427","OS_TOP_HLDG_NAME(ALL,12/31/2017,MTD,ALL,EQ)")`

* *Country* column: `=FDS("M4032427","OS_TOP_HLDG_CNTRY(ALL,12/31/2017,MTD,ALL,EQ)")`

* *CUSIP* column: `=FDS("M4032427","OS_TOP_HLDG_ID(ALL,12/31/2017,MTD,ALL,EQ)")`

* *Market Value USD* column: `=FDS("M4032427","OS_TOP_HLDG_MV(ALL,12/31/2017,MTD,ALL,USD,EQ)")`

7. SDC Data

- *Paths:*

1. `raw/SDC/bonds/[SubsamplePeriod].[xls/xlsx]`
2. `raw/SDC/loans/[SubsamplePeriod].[xls/xlsx]`
3. `raw/SDC/equities/Equities_[SubsamplePeriod].[xls/xlsx]`

- *Description:* These files constitute a bulk download of the Refinitiv SDC Platinum New Issues database (refinitiv.com), and were downloaded directly using the SDC Platinum software platform. The data were downloaded separately for the three sections of the New Issues database: “All Equity”, “All Bonds”, and “All Syndicated Loans”. The data were downloaded in a series of files that cover separate time periods (for example, in half-year increments for the most recent bond data) — hence the notation `[SubsamplePeriod]` in the path structures highlighted above. The particular subsample partitioning is however not essential, and the data could be downloaded using any partitioning. We have included in the replication package mock versions of some of these files with sample rows only, to illustrate their structure.

8. Capital IQ Data

- *Paths:*

1. `raw/ciq/wrds_cusip.dta`
2. `raw/ciq/ciq_ultimate_parents.csv`

- *Description:*

- These files contain the data from Capital IQ (capitaliq.com) that is used in the aggregation procedure. We outline below how each of the files can be obtained.
- File (1) is a list of all the CUSIP identifiers in Capital IQ (CIQ), mapped to the corresponding CIQ internal identifiers. This file can be downloaded directly from the WRDS (wrds-web.wharton.upenn.edu) Unix server. It comes in SAS format and should be converted to Stata format using software such as StatTransfer.
- File (2) provides a link from each company in file (1) to its ultimate parent, inclusive of the parent’s primary CUSIP, CIQID, and country of incorporation. A single-row sample version of the file with mock data is included in the replication packet. Ultimate parents and their CIQ IDs were downloaded for all the CUSIP-linked CIQIDs via CIQ’s excel plug-in. Specifically in the CIQ excel plug-in, the formula to retrieve/download the ultimate parent’s name is `=CIQ($A2, “IQ_ULT_PARENT”)` wither \$A2 refers to the cell where the entity’s CIQID lives. The formula for the ultimate parent’s CIQID is `=CIQ($A2, “IQ_ULT_PARENT_CIQID”)`. We have included in the replication package a mock version of this file with sample rows only, to illustrate its structure.

9. Bureau van Dijk Data

- *Paths:*

1. raw/orbis/header/BvDIDChange.dta
2. raw/orbis/header/ISIN_BvDID.dta
3. raw/orbis/header/LEI_details.dta
4. raw/orbis/ownership_data/[CountryCode]/SHARE_[CountryCode]_Link_all yrs.dta

- *Description:*

- These files contain corporate ownership data from the Orbis database provided by Bureau van Dijk (Bvd). The Orbis data can be sourced directly from BvD (bvdinfo.com). We have included in the replication package mock versions of some of these files with sample rows only, to illustrate their structure.
- Files (1) through (3) contain reference header data. File (1) contains a history of changes in the internal identifiers used by BvD. File (2) contains a mapping from BvD ID to ISIN, and file (3) contains a mapping from BvD ID to Legal Entity Identifier (LEI).
- Item (4) shows the path structure for the Orbis ownership files. Under the path raw/orbis/ownership_data, there should be one folder for each of the countries included in Orbis (the placeholder [CountryCode] above refers to ISO2 codes). The SHARE_* files in these folders contain all historical subsidiary-shareholder links for which the subsidiary is located in the country designated by [CountryCode].

10. Bloomberg Data

- *Paths:*

1. raw/Bloomberg/vie_bloomberg.xlsx

- *Description:* This file contains information on a set of dynamic fields (market capitalization and equity price) and static fields (IPO date, IPO price, and shares offered at IPO) for China's VIE companies. This data was downloaded from a Bloomberg terminal via Bloomberg's Excel API. We include in the replication package an example Excel workbook that outlines the required structure and the API formulas that we used. The workbook contains three sheets. The first sheet, called "static", has one row per CUSIP, and we include a single example row demonstrating the formulas. The rest of the rows should be populated with the set of equity CUSIPs in the CGS data for which the CMNS aggregation file (cmns_aggregation.dta, produced by the code) assigns residency in the Cayman Islands and nationality in China. The second and third sheets, called "market_cap" and "price", have one column per CUSIP, and the columns should be filled out in the same way: we include example columns demonstrating the formulas and structure of the data.

11. Securities Holdings Statistics (SHS) Data [*]

- *Paths:*
 1. raw/shs/SHS - EMU Holdings of CYP Common Equity.xlsx
 2. raw/shs/SHS - EMU Holdings of CYP Fund Shares.xlsx
 3. raw/shs/SHS - EMU Holdings of IRL Common Equity.xlsx
 4. raw/shs/SHS - EMU Holdings of IRL Fund Shares.xlsx
 5. raw/shs/SHS - EMU Holdings of NLD Common Equity.xlsx
 6. raw/shs/SHS - EMU Holdings of NLD Fund Shares.xlsx
- *Description:* These files were downloaded from the European Central Banks's Securities Holdings Statistics (SHS) database, and report the holdings of Eurozone countries in common equities and fund shares in Cyprus, Ireland, and the Netherlands.

12. CPIS Data [*]

- *Paths:* All files included in the replication packet at `raw/CPIS`.
- *Description:* This dataset is a bulk download of the CPIS time series data for the external assets and external liabilities of many IMF member countries. The data was obtained from the IMF's website.

13. CDIS Data [*]

- *Paths:* All files included in the replication packet at `raw/CDIS`.
- *Description:* This dataset is a bulk download of the CDIS time series data for the external assets and external liabilities of many IMF member countries. The data was obtained from the IMF's website.

14. TIC Data [*]

- *Paths:* All files included in the replication packet at `raw/TIC`.
- *Description:* These files are copies of the statistical reports provided by the U.S. Treasury's Treasury International Capital (TIC) system. Some of the files included in the replication packet have been digitized from PDF copies of TIC's reports. These data provide a statistical account of U.S. external assets and external liabilities. The files include various annual-frequency vintages of the TIC tables used in the paper.

15. SAFE Data [*]

- *Paths:* `raw/SAFE/SAFE_IIP.xlsx`
- *Description:* This file provides a time series for the various categories of China's officially reported international investment position. The data are publicly available from China's State Administration of Foreign Exchange (safe.gov.cn).

16. OpenFIGI API Data

- *Paths:*

1. `raw/figi/figi_master_compact.dta`

- *Description:* This file contains information from Bloomberg’s OpenFIGI API data (including market sector, FIGI identifiers, and other security- and issuer-level information) that is obtained by querying the OpenFIGI API for each individual CUSIP and ISIN code found in the universe of the CUSIP Global Services (CGS) data. Since the OpenFIGI API one-time download script takes several weeks to run for the universe of CUSIPs from CUSIP Global Services (CGS), it is included as a separate stand-alone utility in the replication package (`FIGI_Download.py`), and the corresponding output is treated here as raw data. Running the OpenFIGI download code also requires an API key, which should be obtained directly from OpenFIGI (openfigi.com).²

17. Worldscope Data

- *Paths:*

1. `raw/worldscope/wsndata.dta`
2. `raw/worldscope/wrds_ws_stock.dta`
3. `raw/worldscope/wrds_ws_fundamentals_annual.dta`
4. `raw/worldscope/ws_countries.dta`

- *Description:* All files correspond to tables in Refinitiv’s Worldscope database, which can be obtained from WRDS (wrds.wharton.upenn.edu). File 1 corresponds to the WSNDATA table in Worldscope (see here for a listing of the database tables). File 2 corresponds to the “Stock Data” table. File 3 corresponds to the “Fundamentals Annual” table. File 4 corresponds to the WSCURR table.

18. TRACE Data

- *Paths:*

1. `raw/wrds/trace_master/trace_master_corp_agency.dta`
2. `raw/wrds/trace_master/trace_master_abs.dta`
3. `raw/wrds/trace_master/trace_master_cmo.dta`
4. `raw/wrds/trace_master/trace_master_mbs.dta`
5. `raw/wrds/trace_master/trace_master_tba.dta`
6. `raw/wrds/wrds_bond_returns.dta`

- *Description:* All files correspond to tables in TRACE and in the WRDS bond returns database (derived from TRACE), which can be obtained from WRDS (wrds.wharton.upenn.edu). This data is used in the build of our security masterfile.

²The code also references versions of the `figi_master_compact` file with suffixes `_cusip_unique` and `_isin_unique`. These are simply subsets of the `figi_master_compact` file where records with missing ISIN codes are dropped (`_cusip_unique`), or records with missing CUSIP codes are dropped (`_isin_unique`).

19. IMF Data [*]

- *Paths:* All included data under `raw/macro/concordances` and `raw/macro/IMF_IFS`.
- *Description:* This data was sourced from the IMF’s website and provides exchange rate data from the IMF’s International Financial Statistics (IFS) data, as well as basic country- and currency-level symbology files.

20. World Bank Data [*]

- *Paths:* All data included in the replication packet at `raw/World_Bank`.
- *Description:* This data was sourced from the World Bank’s website and provides current account and GDP data at the country level.

21. Other Utilities [*]

- *Paths:*
 1. `raw/Internal/Security-Masterfile-Dictionary.xlsx`
- *Description:*
 - File (1) provides a specification of the asset class hierarchy used in the internal security masterfile, and also includes crosswalks from external asset class classifications to the internal one.

References

- Coppola, Antonio, Matteo Maggiori, Brent Neiman, and Jesse Schreger**, “Redrawing the Map of Global Capital Flows: The Role of Cross-Border Financing and Tax Havens,” *Forthcoming at the Quarterly Journal of Economics*, 2021.
- Maggiori, Matteo, Brent Neiman, and Jesse Schreger**, “International currencies and capital allocation,” *Journal of Political Economy*, 2020, 128 (6), 2019–2066.